Genomic & Pharmacogenomic Clinical Summary

Patient / Sample ID: Melinda Ehrle (Melinda_Ehrle) | **Assay:** 33.4x BAM mean depth Phased WGS (GRCh38.p14) | **Date:** October 12, 2026

MONOGENIC CARRIERS
4 (Het)
POLG, TAT, ATM (Recessive/Dom)

PHARMACOGENOMICS
9 Priority
ABCG2, CYP2C19, CYP2W1, CYP3A43

DISEASE MODIFIERS
3 Modifiers
Organ / Metabolic

1. PRIORITY GENOMIC FINDINGS & CARRIER STATUS

Primary Care & Specialist Triage

GENE	VARIANT & LOCUS	ZYGOSITY	CLASSIFICATION	AI IMPACT (PHRED)	CLINICAL IMPACT & PRIMARY CARE GUIDANCE
POLG	p.Gly737Arg · c.2209G>C · rs121918054	Heterozygous	Pathogenic	AVI 30.3 / CADD 26.3 / REV 0.936	Pathogenic missense variant (p.Gly737Arg) in the catalytic subunit of mitochondrial DNA polymerase gamma (OMIM #174763), essential for mitochondrial genome replication. **Clinical Impact:** Heterozygous carriers are typically asymptomatic at baseline, but carry a **strict FDA Boxed Warning contraindication against valproic acid / divalproex sodium** due to risk of fatal valproate-induced hepatic necrosis; verify baseline AST/ALT on annual metabolic panels.
TAT	p.Arg57Ter · c.169C>T · rs118203914	Heterozygous	Pathogenic	AVI 38.8 / CADD 36.0	Premature stop-gained variant (p.Arg57Ter) in hepatic cytosolic tyrosine aminotransferase, the rate-limiting enzyme converting L-tyrosine to p-hydroxyphenylpyruvate, associated with autosomal recessive **Tyrosinemia Type II (Richner-Hanhart syndrome, OMIM #276600)**. **Clinical Impact:** Heterozygous carriers retain ~50% hepatic enzymatic activity via the wild-type allele, remain clinically asymptomatic on an unrestricted diet, and do **not** develop oculocutaneous keratitis or palmoplantar hyperkeratosis (no dietary tyrosine/phenylalanine restriction indicated; optional one-time fasting plasma tyrosine < 100 µmol/L confirms normal clearance).
ATM	c.1236-2del · rs1565381646	Heterozygous	Pathogenic	Analytical VCF	Pathogenic canonical splice-acceptor deletion (c.1236-2del) in the ATM serine/threonine DNA double-strand break repair kinase (OMIM #114480). **Clinical Impact:** Heterozygous female carriers have a moderately elevated lifetime breast cancer risk (~20–30%) and increased pancreatic cancer risk; NCCN guidelines recommend annual breast MRI (starting age 30–35) and annual mammography, family history review for pancreatic screening, and minimizing unnecessary diagnostic ionizing radiation.
VDR	c.-1172A>G · rs4516035	Heterozygous	Pathogenic	AVI 2.1 / CADD 5.4	5'-UTR promoter regulatory variant (c.-1172A>G) in the nuclear vitamin D3 (calcitriol) receptor (OMIM #601769), which regulates intestinal calcium transport and osteoblast gene transcription. **Clinical Impact:** Modulates baseline calcitriol receptor expression; maintain serum 25(OH)D at 35–55 ng/mL via routine

Prescribing Guardrails & Annual Preventive Surveillance Schedule

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2. PRESCRIBING & PERIOPERATIVE GUARDRAILS (CPIC LEVEL A & ACUTE RISK TRIAGE)

GENE & VARIANT / DIPLOTYPE	PHENOTYPE / MECHANISM	HIGH-RISK MEDICATIONS / EXPOSURES	PRESCRIBING ACTION & GUIDANCE (CPIC LEVEL A GROUNDED)
POLG p.Gly737Arg · c.2209G>C · rs121918054	Mitochondrial Carrier	⚠ LIFE-THREATENING RISK **Valproic Acid / Sodium Valproate**	**ABSOLUTE CONTRAINDICATION (FATAL RISK):** High risk of fatal valproate-induced hepatic necrosis (FDA Boxed Warning / CPIC). Strictly avoid valproate formulations.
ABCG2 *ref/*rs2231142 (Q141K (rs2231142) G>T)	Decreased Function	**Allopurinol, Clozapine, Imatinib, Methotrexate**	**DOSE_DOWN:** Allele G is associated with decreased dose of allopurinol or febuxostat in people with Gout as compared to allele A.
CYP2C19 *2/*2 (splice defect (rs12769205) A>G, splice defect (rs4244285) G>A)	Poor Metabolizer	⚠ LIFE-THREATENING RISK **Clopidogrel (Plavix)**	**AVOID / CONTRAINDICATION:** Significantly reduced active metabolite formation and high risk of cardiovascular ischemic events/stent thrombosis. Avoid clopidogrel; prescribe alternative antiplatelet agent (Prasugrel or Ticagrelor).
CYP2W1 *6/*6 (P488L (rs3808348) C>T)	Unknown	**CPIC Substrates**	**MONITOR:** Titrate dosing per CPIC.
CYP3A43 *2/*3 (Y25fs (rs61469810) delA, P340A (rs680055) C>G)	Unknown	**CPIC Substrates**	**MONITOR:** Titrate dosing per CPIC.
CYP3A5 *3/*3 (splice defect (rs776746) T>C)	Poor Metabolizer	**Tacrolimus (Prograf)**	**STANDARD:** CYP3A5 *3/*3 non-expressor (Poor Metabolizer). Requires standard lower initial dosing (0.10-0.15 mg/kg/day) compared to CYP3A5 expressors to achieve target trough concentrations. Monitor blood levels (TDM).
CYP4F2 *2/*3 (V433M (rs2108622) C>T, W12G (rs3093105) A>C)	Altered Vitamin K / Warfarin Metabolism	**CPIC Substrates**	**MONITOR:** Titrate dosing per CPIC.
NAT2 *4/*7 (R268K (rs1208) G>A)	Intermediate Metabolizer	**CPIC Substrates**	**MONITOR:** Titrate dosing per CPIC.

3. SUGGESTED CLINICAL & LABORATORY SURVEILLANCE SCHEDULE (ANNUAL PHYSICAL)

EVALUATION / LABORATORY TEST	TRIGGERING GENE, VARIANT & ZYGOSITY	SUGGESTED FREQUENCY	CLINICAL RATIONALE & ACTIONABLE THRESHOLDS
Comprehensive Metabolic Panel (CMP — Hepatic AST, ALT, Bilirubin & eGFR) Venous Blood Draw + Permanent EHR Valproate Allergy Flag	POLG p.Gly737Arg · c.2209G>C · rs121918054 chr15:89323460 · DP: 31x Heterozygous (Autosomal Dominant)	Annual PCP Preventive Blood Panel & permanent EHR prescribing alert	Pathogenic missense variant (p.Gly737Arg) in the catalytic subunit of mitochondrial DNA polymerase gamma (OMIM #174763), essential for mitochondrial genome replication. Clinical Impact: Heterozygous carriers are typically asymptomatic at baseline, but carry a strict FDA Boxed Warning contraindication against valproic acid / divalproex sodium due to risk of fatal valproate-induced hepatic necrosis; verify baseline AST/ALT on annual metabolic panels.
Fasting Plasma Tyrosine / Amino Acid Profile & Hepatic AST/ALT Optional Baseline Venous Blood Draw	TAT p.Arg57Ter · c.169C>T · rs118203914 chr16:71576247 · DP: 52x Heterozygous (Autosomal recessive)	Optional one-time baseline at routine PCP physical	Premature stop-gained variant (p.Arg57Ter) in hepatic cytosolic tyrosine aminotransferase, the rate-limiting enzyme converting L-tyrosine to p-hydroxyphenylpyruvate, associated with autosomal recessive Tyrosinemia Type II (Richner-Hanhart

Clinical Decision Calculus, Evidence Confidence & EHR Integration

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4. CLINICAL DECISION CALCULUS: BALANCED INTERVENTIONAL RATIONALE

✅ ARGUMENTS FOR TARGETED PREVENTIVE SURVEILLANCE & PGX

- **Immediate Medication Safety:** Documenting CPIC pharmacogenomic diplotypes prevents avoidable drug toxicity and therapeutic failure before the first prescription is filled.
- **Low-Cost Biomarker Baselines:** Pathway-matched venous blood tests incorporate seamlessly into the annual physical without procedural diagnostic burdens.
- **Proactive Specialist Coordination:** Sharing variant-specific guardrails ensures appropriate situational precautions without over-medicalization.

🛑 ARGUMENTS AGAINST AGGRESSIVE OVER-INTERVENTION

- **Heterozygous Carrier & Modifier Reserve:** Heterozygous variants retain an intact wild-type parental allele (~50–100% capacity) and **do not warrant invasive diagnostic procedures**.
- **Phased Allelic Reassurance:** Whole-genome parental phasing confirms **0 confirmed pathogenic biallelic (trans) recessive compound-heterozygotes**.
- **Polygenic Modifiers ≠ Deterministic Disease:** Modest statistical variants must never trigger unprovoked chronic anticoagulation or off-label polypharmacy.

5. CALCULATED EVIDENCE CONFIDENCE METRICS

EVIDENCE DOMAIN	CALCULATED CONFIDENCE	ANALYTICAL BASIS & CONCORDANCE FACTORS
WGS Analytical Callset	**98%**	33.4x BAM mean depth pan-genome aligned short-read WGS (GRCh38.p14).
Primary Pathogenic Carrier Status (POLG, TAT, ATM)	**95%**	Orthogonal concordance between ClinVar Pathogenic curations and AlphaGenome AVI / CADD scores.
Pharmacogenomic PGx Diplotypes (ABCG2, CYP2C19, CYP2W1)	**98%**	Dual-engine verified concordance (PharmCAT + Aldy consensus); CPIC Level A clinical guidelines.
Endogenous Protective & Polygenic Risk Alleles	**65%**	Population-level statistical GWAS meta-analyses requiring longitudinal correlation.

6. RECOMMENDED CLINICAL NOTE / EHR PROBLEM-LIST INTEGRATION TEXT

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PATIENT GENOMIC REVIEW SUMMARY (33.4x BAM Mean Depth Phased WGS):
- KEY GENOMIC & CARRIER FINDINGS: POLG (p.Gly737Arg · c.2209G>C · rs121918054, Heterozygous); TAT (p.Arg57Ter · c.169C>T · rs118203914, Heterozygous); ATM (c.1236-2del · rs1565381646, Heterozygous); VDR (c.-1172A>G · rs4516035, Heterozygous); MC1R (p.Asp294His · c.880G>C · rs1805009, Heterozygous). Zero biallelic pathogenic recessive compound-heterozygotes detected; heterozygous carrier/modifier findings retain one wild-type allele and do NOT warrant invasive diagnostic procedures.
- PHARMACOGENOMIC GUARDRAILS (CPIC Level A): ABCG2 *ref/*rs2231142. Review patient CPIC diplotypes prior to prescribing; adhere to dosage reduction, drug avoidance, and TDM precautions.
- PREVENTIVE SURVEILLANCE: Incorporate pathway-matched venous blood panels and preventive wellness checks during the routine annual physical.
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